



**THOUGHTSTM
TO PAPER**



COMPREHENSIVE PATENT SEARCH REPORT

March 13, 2026

Invention: CLOOMC
Case Number: TPS98134

Kenneth J Hamer-Hodges:

A patent search has been conducted and a report prepared. A patent search report discovers inventions similar to yours. Each similar invention found is called “prior art” and are all inventions disclosed before yours. To be granted a patent by the US Patent & Trademark Office (USPTO), a Patent Office examiner must determine that your invention meets the three patenting criteria of being novel, non-obvious, and useful when compared to prior art.

Having a patent search report provides you with valuable information. The report will show you prior art that may be cited by the USPTO when they look at a patent application for your invention. Using this information, you can make a preliminary assessment towards whether or not the USPTO will consider your invention patentable. You can then decide if you wish to continue to file a patent application. In addition, you can make changes or improvements to your invention to make your invention more unique and therefore more patentable, before submitting a patent application.

Upon reviewing this search report, determine whether you can make improvements to your invention to increase the uniqueness of your invention above the prior art. Send any such improvements to us before deciding to proceed with a patent application. If you have any questions and wish to discuss the search report, please let us know. We will wait for your decision to proceed with a patent application.

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**NO GUARANTEE OF FINDING ALL SIMILAR INVENTIONS OR FUTURE PATENT APPROVAL.
NOT AN INFRINGEMENT OPINION OR FREEDOM TO OPERATE OPINION**

You understand and accept that there is absolutely no guarantee that we will find all similar inventions (prior art), or that we will find all prior art that may be used by a patent examiner to potentially reject your patent application should you decide to file a patent application in the future. Whether a prior art is considered similar enough to your invention is a subjective determination which will vary from individual to individual. Therefore, it is possible that we may find prior art that a patent examiner will not use in a patent application rejection or conversely, a patent examiner may use a prior art in a patent application rejection that we do not find. Patent searching is not an exact science and it is common for prior art to not be found. Further, recently filed patent applications not published cannot be found in a patent search. A patent search should not be relied upon as a definitive conclusion as to whether your invention can or cannot obtain a patent. Instead, a patent search should be used as a general assessment of the patentability of your invention, as well as for use in improving your invention over existing inventions to increase your overall patentability. Any patentability opinion that is provided in the search report is a professional opinion only and comes with absolutely no guarantee of the outcome of a patent application. You agree to hold us harmless for any actions or inactions you may take while relying on the search report we provide. Should a patent application be rejected, there is no refund of any fees paid, including this patent search fee. You agree not to seek losses from us for any costs you may have incurred through reliance of our patent search.

A patent search report gives a general assessment as to the likelihood an invention can obtain a patent. It does not give an opinion as to you have the freedom to sell your invention without infringing upon another's patent. It is possible for your invention to be patentable and obtain a patent, and yet you cannot make the invention without licensing parts of the invention from another patent holder. A patent search discusses patentability only, the likelihood you can obtain a patent.

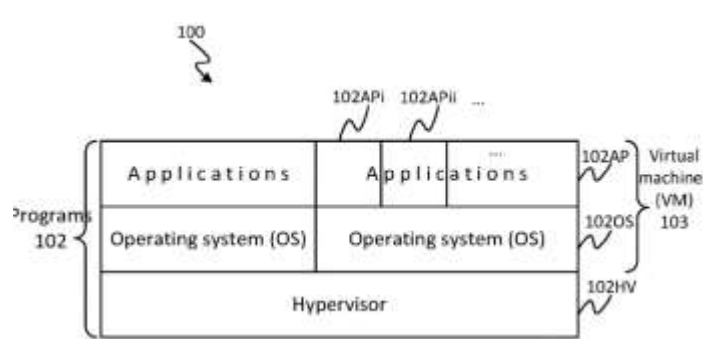
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PRIOR ART

Prior art which is relevant to your invention is listed in the table below. For each prior art, a rank number is assigned which represents how important the particular prior art is to your invention. A smaller rank number represents a prior art that is more relevant than a prior art with a larger rank number. Full copies of the prior art listed are in the Prior Art Portfolio, a separate file or document.

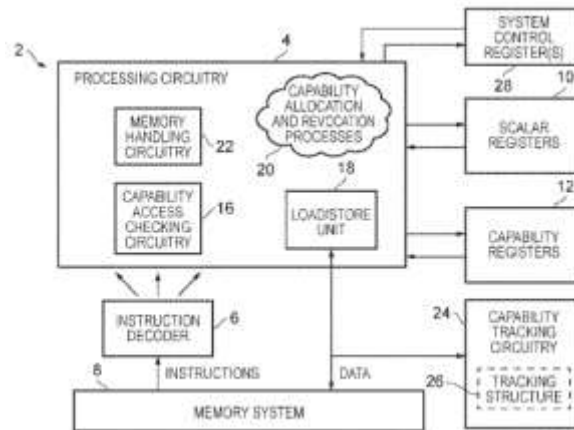
When browsing through the prior art, it will be beneficial to write down how your invention is different and how it may overcome the prior art's shortcomings. Including this information will benefit your patent application.

Rank	Document Number	Invention Title / Similarities Noted & Comments
3	US12061541B2	System for confidential computing with capabilities The reference describes hardware capability checks triggered by load/store memory-access instructions and also describes non-retained ("linear") capability transfer, covering dual-path access checks and a form of propagation restriction. Not disclosed: It does not describe a per-entry bind bit in a namespace/capability list that is checked specifically on a capability-list write operation to permit or deny propagation into another list.  [Abstract] According to a first aspect, execution logic is configured to perform a linear capability transfer operation which transfers a physical capability from a partition of a first software modules to a partition of a second of software module without retaining it in the partition of the first. According to a second, alternative or additional aspect, the execution logic is configured to perform a sharding operation whereby a physical capability is divided into at least two instances, which may later be combined. [Relevant Excerpts] [Summary] The present disclosure relates to a system comprising: execution logic comprising one or more execution units, memory comprising one or more memory units, and memory access logic for looking-up a mapping between a physical address space of the memory and a virtual address space. The execution logic is arranged to run a plurality of software modules each in a respective partition of the physical address space, including to execute a memory access operation performed by a currently-executing one of said software modules. The memory access operation specifies a virtual address in the virtual address range as a target of the memory access operation. The memory access logic is configured to perform operations of: looking-up a physical address in the physical address space mapped to the virtual address,

		checking whether a capability granting access to said physical address is found in a storage location to which the currently-executing software module has access, and on condition thereof, granting access to said physical address for performing said memory access operation. According to one aspect disclosed herein, the execution logic is configured to perform a linear capability transfer operation which transfers one of the capabilities from the partition of a first of said software modules to the partition of a second of said software modules without retaining it in the partition of the first. According to a second aspect disclosed herein, the execution unit is configured to perform a sharding operation whereby one of the capabilities is divided into at least two instances, and the execution unit is further configured to perform a shard recombination operation whereby the instances are recombined. The first and second aspects may be used together or independently. This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Nor is the claimed subject matter limited to implementations that solve any or all of the disadvantages noted herein.
3	WO2023067295A1	Technique for handling sealed capabilities The reference describes sealed capabilities used as callback tokens and enforced restrictions on use based on trust level, disclosing token-like capability handling and controlled use/propagation semantics. Not disclosed: It does not describe a capability-based lambda-calculus function-application instruction that performs an in-domain call and stores a return address into a machine-status register. [Abstract] An apparatus and method are described for handling sealed capabilities. The apparatus has processing circuitry to perform processing operations during which access requests to memory are generated, wherein the processing circuitry is arranged to generate memory addresses for the access requests using capabilities

		that identify constraining information. Checking circuitry then determines whether a given access request whose memory address is generated using a given capability is permitted based on the constraining information identified by that given capability, and based on a level of trust associated with the given access request. Each capability has a capability level of trust associated therewith, and the level of trust associated with the given access request is dependent on both a current mode level of trust associated with a current mode of operation of the processing circuitry, and the capability level of trust of the given capability. At least one of the capabilities is settable as a sealed capability, and the apparatus further comprises sealed capability handling circuitry to prevent the processing circuitry performing at least one processing operation using a given sealed capability when the current mode level of trust is a lower level of trust than the capability level of trust of the given sealed capability. [Relevant Excerpts] [Summary] In a first example arrangement, there is provided an apparatus comprising: processing circuitry to perform processing operations during which access requests to memory are generated, wherein the processing circuitry is arranged to generate memory addresses for the access requests using capabilities that identify constraining information; and checking circuitry to determine whether a given access request whose memory address is generated using a given capability is permitted based on the constraining information identified by that given capability, and based on a level of trust associated with the given access request; wherein: each capability has a capability level of trust associated therewith, and the level of trust associated with the given access request is dependent on both a current mode level of trust associated with a current mode of operation of the processing circuitry, and the capability level of trust of the given capability; at least one of the capabilities is settable as a sealed capability; and the apparatus further comprises sealed capability handling circuitry to prevent the processing circuitry performing at least one processing operation using a given sealed capability when the current mode level of trust is a lower level of trust than the capability level of trust of the given sealed capability. In a further example arrangement, there is provided a method of handling sealed capabilities, comprising: employing processing circuitry to perform processing operations during which access requests to memory are generated, wherein the processing circuitry generates memory addresses for the access requests using capabilities that identify constraining information; and determining whether a given access request whose memory address is generated using a given capability is permitted based on the constraining information identified by that given capability, and based on a level of trust associated with the given access request; wherein: each capability has a capability level of trust associated therewith, and the level of trust associated with the given access request is dependent on both a current mode level of trust associated with a current mode of operation of the processing circuitry, and the capability level of trust of the given capability; and at least one of the capabilities is settable as a sealed capability; and the method further comprises employing sealed capability handling circuitry to prevent the processing circuitry performing at least one processing operation using a given sealed capability when the current mode level of trust is a lower level of trust than the capability level of trust of the given sealed capability.
3	WO2023203307A1	Apparatus and method for managing capabilities The reference teaches write-side tracking when a capability is written into a memory region (setting a tracking field), aligning with monitoring/controlling capability propagation via stores.

Not disclosed: It does not describe a propagation-permit/bind bit that is checked at the time of writing a capability into a distinct capability-list namespace to allow or deny copying into that list.



[Abstract]

An apparatus is described having processing circuitry for performing operations during which access requests to memory are generated. The processing circuitry generates memory addresses for the access requests using capabilities, where each capability indicates a pointer value and constraining information used to constrain access to memory using memory addresses derived from the pointer value. A marker indication field is stored in association with each capability to provide a marker value used to distinguish between static capabilities used to access statically allocated memory and dynamic capabilities used to access dynamically allocated memory. Capability tracking circuitry maintains a tracking structure providing a tracking field for each of a plurality of memory regions, and the capability tracking circuitry sets the tracking field for a given memory region amongst the plurality of memory regions when at least one capability whose associated marker indication field has a specified marker value is written to the given memory region. The specified marker value indicates that writing of the associated capability to memory is to be tracked by the capability tracking circuitry to facilitate subsequent revocation of that associated capability.

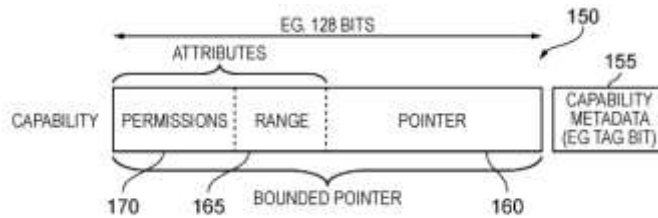
[Relevant Excerpts]

[Summary]

In one example arrangement, there is provided an apparatus comprising: processing circuitry to perform operations during which access requests to memory are generated, wherein the processing circuitry is arranged to generate memory addresses for the access requests using capabilities, where each capability is arranged to indicate a pointer value and constraining information used to constrain access to memory using memory addresses derived from the pointer value, and wherein a marker indication field is stored in association with each capability to provide a marker value used to distinguish between static capabilities used to access statically allocated memory and dynamic capabilities used to access dynamically allocated memory; and capability tracking circuitry to maintain a tracking structure providing a tracking field for each of a plurality of memory regions, wherein the capability tracking circuitry is arranged to set the tracking field for a given memory region amongst the plurality of memory regions when at least one capability whose associated marker indication field has a specified marker value is written to the given

		memory region, where the specified marker value indicates that writing of the associated capability to memory is to be tracked by the capability tracking circuitry to facilitate subsequent revocation of that associated capability. In another example arrangement, there is provided a method of operating an apparatus to track capabilities for revocation, comprising: employing processing circuitry to perform operations during which access requests to memory are generated, wherein the processing circuitry generates memory addresses for the access requests using capabilities, where each capability is arranged to indicate a pointer value and constraining information used to constrain access to memory using memory addresses derived from the pointer value; storing a marker indication field in association with each capability to provide a marker value used to distinguish between static capabilities used to access statically allocated memory and dynamic capabilities used to access dynamically allocated memory; maintaining, using capability tracking circuitry, a tracking structure providing a tracking field for each of a plurality of memory regions; and setting the tracking field for a given memory region amongst the plurality of memory regions when at least one capability whose associated marker indication field has a specified marker value is written to the given memory region, where the specified marker value indicates that writing of the associated capability to memory is to be tracked by the capability tracking circuitry to facilitate subsequent revocation of that associated capability.
3	WO2018215734A1	An apparatus and method for interpreting permissions associated with a capability The reference discloses capability permission-flag interpretation mechanisms enabling additional permission semantics from existing capability bits, consistent with implementing a dedicated propagation-control permission within capability metadata. Not disclosed: It does not describe a specific “bind/non-propagate” bit enforced on capability-list write operations to prevent propagation between capability lists or namespaces. [Abstract] An apparatus and method are provided for interpreting permissions associated with a capability. The apparatus has processing circuitry for executing instructions in order to perform operations, and a capability storage element accessible to the processing

		circuitry and arranged to store a capability used to constrain at least one operation performed by the processing circuitry when executing the instructions. The capability identifies a plurality N of default permissions whose state, in accordance with a default interpretation, is determined from N permission flags provided in the capability. In accordance with the default interpretation, each permission flag is associated with one of the default permissions. The processing circuitry is then arranged to analyse the capability in accordance with an alternative interpretation, in order to derive, from logical combinations of the N permission flags, state for an enlarged set of permissions, where the enlarged set comprises at least N+1 permissions. This provides a mechanism for encoding additional permissions into capabilities without increasing the number of permission flags required, whilst still retaining desirable behaviour. [Relevant Excerpts] [Page 1, Line 5-26] There is increasing interest in capability-based architectures in which certain capabilities are defined for a given process, and an error can be triggered if there is an attempt to carry out operations outside the defined capabilities. The capabilities can take a variety of forms, but one type of capability is a bounded pointer (which may also be referred to as a "fat pointer"). A number of capability storage elements (for example registers) may be provided for storing capabilities for access by the processing circuitry of a data processing apparatus. Each capability can include a number of permission flags identifying one or more permissions associated with the capability, for example to restrict the use of the capability to certain types of operation, certain modes of operation of the processing circuitry, etc. For instance, considering a bounded pointer within such a capability storage element, this may identify a non-extendible range of memory addresses accessible by the processing circuitry, along with one or more permission flags identifying associated permissions. Whilst the processing circuitry may be allowed to take steps to reduce the range and/or clear permission flags associated with any particular bounded pointer available to it, it cannot in normal operation extend the range, or set permission flags, in order to seek to increase the capability afforded to the processing circuitry by that bounded pointer. It would be desirable to allow the number of permissions associated with a capability to be extended, but the number of bits used to encode a capability is typically constrained, for example by the size of the capability storage elements in which the capabilities are stored.
3	WO2017182770A1	An apparatus and method for performing operations on capability metadata The reference teaches explicit operations that access and modify capability metadata (including setting/clearing metadata identifying a data block as a capability), relevant to write-path validation/markings of capability-bearing storage. Not disclosed: It does not describe a dual read-gate/write-gate mechanism that validates a capability token's internal fields on every capability-list read and capability-list write.



[Abstract]

An apparatus is provided comprising storage elements to store data blocks, where each data block has capability metadata associated therewith identifying whether the data block specifies a capability, at least one capability type being a bounded pointer. Processing circuitry is then arranged to be responsive to a bulk capability metadata operation identifying a plurality of the storage elements, to perform an operation on the capability metadata associated with each data block stored in the plurality of storage elements. Via a single specified operation, this hence enables query and/or modification operations to be performed on multiple items of capability metadata, hence providing more efficient access to such capability metadata.

[Relevant Excerpts]

[Page 1, Line 4-27]

The present technique relates to an apparatus and method for performing operations on capability metadata.

There is increasing interest in capability-based architectures in which certain capabilities are defined for a given process, and an error can be triggered if there is an attempt to carry out operations outside the defined capabilities. The capabilities can take a variety of forms, but one type of capability is a bounded pointer (which may also be referred to as a "fat pointer"). For a bounded pointer, the pointer value may identify, or be used to determine, the address of a data value to be accessed or an instruction to be executed, for example. However, the pointer value may also have associated range information which indicates an allowable range of addresses when using the pointer value. This can be useful for example for ensuring that the address determined from the pointer remains within certain bounds to maintain security or functional correctness of behaviour. In addition, certain permission/restriction information may be specified in association with the pointer value of a bounded pointer. The range information and any permissions/restrictions information for a bounded pointer may be referred to as capability information, and within a capability-based architecture such a bounded pointer (including its associated capability information) may be referred to as a capability.

Within a capability-based architecture, it is known to store capability metadata in association with each data block stored within storage elements of the apparatus. The capability metadata can be used to identify whether the associated data block specifies a capability, or instead contains data that does not represent a capability (also referred to herein as general purpose data). If desired, the capability metadata can also specify certain additional information.

3

[Reference 1](#)

Capability Hardware Enhanced RISC Instructions: CHERI Instruction-Set Architecture (Version 9)

The reference defines permission-checked capability loads/stores (including

		constraints on storing capabilities) and defines CJALR as a capability jump-and-link primitive that records a return capability, matching dual-path validation and capability-based call/link behavior. Not disclosed: It does not describe a lambda-calculus-named instruction that performs function application while storing the return address specifically in a machine-status register. [Abstract] This technical report describes CHERI ISAv9, the ninth version of the CHERI architecture being developed by SRI International and the University of Cambridge. This design captures thirteen years of research, development, experimentation, refinement, formal analysis, and validation through hardware and software implementation. CHERI introduces an architecture-neutral capability-based protection model, which has been instantiated in various commodity base architectures to give CHERI-RISC-V, Arm's prototype Morello architecture, and (sketched) CHERI-x86-64. It enables software to efficiently implement fine-grained memory protection and scalable software compartmentalization, by providing strong, deterministic, efficient mechanisms to support the principles of least privilege and intentional use in the execution of software at multiple levels of abstraction, preventing and mitigating vulnerabilities. Design goals include incremental adoptability from current ISAs and software stacks, low performance overhead for memory protection, significant performance improvements for software compartmentalization, formal grounding, and programmer-friendly underpinnings. CHERI blends traditional paged virtual memory with an in-address-space capability model that includes capability values in registers, capability instructions, and tagged memory to enforce capability integrity. This hybrid approach addresses the performance and robustness issues that arise when trying to express more secure, privilege minimising programming models, above conventional architectures that provide only MMU-based protection. CHERI builds on the C-language fat-pointer literature: its capabilities can describe fine-grained regions of memory, and can be substituted for data or code pointers in generated code, protecting data and improving control-flow robustness. Strong capability integrity and monotonicity properties allow CHERI to express a variety of protection idioms, from enforcing valid C-language pointer provenance and bounds checking to implementing the isolation and controlled communication structures required for software compartmentalization. CHERI's hybrid approach allows incremental adoption of capability-oriented design: critical components can be ported and recompiled to use capabilities throughout, providing finegrain memory protection, or be largely unmodified but encapsulated in ways that permit only controlled interaction. Potential early deployment scenarios include low-level software Trusted Computing Bases (TCBs) such as separation kernels, hypervisors, and operating-system kernels, userspace TCBs such as language runtimes and web browsers, and particularly high-risk software libraries such as data compression, protocol parsing, and image processing (which are concentrations of both complex and historically vulnerability-prone code exposed to untrustworthy data sources). CHERI ISAv9 is a substantial enhancement to prior ISA versions. CHERI-RISC-V has replaced CHERI-MIPS as the primary reference platform, and CHERI-MIPS has been removed from the specification. CHERI architectures now always use merged register files where existing general-purpose registers are extended to support capabilities.
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		CHERI architectures have adopted two design decisions from Arm Morello: 1) CHERI architectures now clear tags rather than raising exceptions if an instruction attempts a non-monotonic modification of a capability; and 2) DDC and PCC no longer relocate legacy memory accesses by default. CHERI-RISC-V has received numerous updates to serve as a better baseline for an upstream standard proposal including a more mature definition of compressed instructions in capability mode. CHERI-x86-64 now includes details of extensions to existing x86 instructions and proposed new instructions in a separate ISA reference chapter along with various other updates.
<i>* Any form of public disclosure can be used as prior art, including un-patented inventions, products, and any information found on the World Wide Web.</i>		

UNIQUE ASPECTS OF YOUR INVENTION

These are components or aspects of your invention which are unique when compared to the prior art. These will be used to argue approval of a patent application. Ensure that unique aspects of your invention are clearly explained in detail. It is beneficial to have as many unique aspects as possible. You may be able to improve on the similarities in the table at the top of the page and convert them into a unique aspect for your invention.

- A processor architecture comprising a dual-gate Trusted Security Base including an mLoad read gate and an mSave write gate that perform hardware validation of capability tokens on both read operations and write propagation operations.
- A capability-based access control mechanism in which each namespace entry includes a bind (B) bit that is checked by a write-side gate to control whether a capability token can be propagated to another capability list.
- A processor instruction implementing lambda calculus function application, wherein a LAMBDA instruction executes code referenced by a capability token within the same protection domain while storing a return address in a machine-status register.

UTILITY AND/OR DESIGN SUITABILITY

Utility patents and design patents work in parallel with each other. One is not a replacement for the other. Utility patent protects functionality whereas design patent protects only the exterior shape of a product. For example, Apple has utility patents to protect the iPhone's function but also has design patents to protect the iPhone's exterior body shape.

Below is whether your invention is suitable for utility patent and design patent. If suitable for both, you can decide to file both to protect the function (utility) and exterior shape (design).

SUITABLE FOR UTILITY PATENT

Protects the functionality, components, and how it works.

Yes

SUITABLE FOR DESIGN PATENT

Protects the exterior shape. Does not protect functionality.

No

Your invention includes methods, and is best suitable for utility patent

CHANCES OF APPROVAL FOR UTILITY PATENT

Utility Patent Protects: The invention's functionality, components, and how it works.

Our opinion as to the likelihood your invention will pass each of the three patentability criteria if submitted to the US Patent Office as a non-provisional utility patent application.

NOVEL

Chance of being considered a new idea

Good

NON-OBVIOUS

Chance of being considered not easily thought of by someone in the industry

Good

USEFUL

Chance of being considered useful

Good

PATENTABLE METHOD

Inventions relating to a method for doing something, a business method, or a computer application are put to an additional test and may be considered unpatentable in certain situations. See the section 'Patentability Risk For Method Inventions' for more details.

Good

OVERALL CHANCE OF UTILITY APPROVAL

Good

PATENTABILITY RISK FOR METHOD INVENTIONS

Inventions relating to a method for doing something, a business method, or a computer application may be at risk for being considered unpatentable in certain situations.

Your invention relates to a method for doing something, a business method, or a computer program, which are all broadly considered methods patents. If in the opinion of the US Patent Office, your invention falls into a category of unpatentable methods, your patent application may be rejected. We believe your invention may possibly fall in one of the categories below. However, whether your invention is considered patentable or unpatentable is *subjective* and ultimately up to the US Patent Office. Therefore, you should know that a risk of rejection exists, as it does for filing any patent application, although you have an additional risk when compared to an invention that is not a method. If your patent application is rejected, response arguments can be made to refute a rejection.

Your invention may fall within one of these unpatentable method categories, with the category or categories of most relevance to your invention bolded:

- Basic economic practices or theories (e.g., hedging, insurance, financial transactions, marketing);
- Basic legal theories (e.g., contracts, dispute resolution, rules of law);
- Mathematical concepts (e.g., algorithms, spatial relationships, geometry);
- Mental activity (e.g., forming a judgment, observation, evaluation, or opinion);
- Interpersonal interactions or relationships (e.g., conversing, dating);
- Teaching concepts (e.g., memorization, repetition);
- Human behavior (e.g., exercising, wearing clothing, following rules or instructions);
- Instructing “how business should be conducted.”
- Laws of nature and physical phenomena
- **Generic computer functions that are well-understood, routine, and conventional activities previously known to the industry**
- Methods which are a basic tool of science and technology where a patent would impede innovation more than it would promote it

If the US Patent Office considers your invention to be related to any of the categories above, they may issue a rejection.

Whether you will get a rejection is uncertain until you file a non-provisional full patent application. You should be aware of the added risk of rejection while making your decision to proceed with a patent application relating to method inventions.

ANALYSIS & RECOMMENDATIONS

Our analysis as to the patentability of your invention along with any recommendations to increase the chances of patent application approval.

You have opted for our comprehensive patent search which covers US, European, and International patent databases. In the interest of being thorough and to thank you for choosing to do business with our firm, we have performed a thorough patent search of these databases to give you the best possible search report and analysis of current prior art (previous inventions). This is similar to the process a patent officer at the United States Patent and Trademark Office (USPTO) would use when searching for prior art to reject your patent application. The patent office can use any patent worldwide as evidence to say your invention is not unique enough to be eligible for a patent.

The reference (US12061541B2) describes hardware capability checks triggered by load/store memory-access instructions and also describes non-retained (“linear”) capability transfer, covering dual-path access checks and a form of propagation restriction. Not disclosed: It does not describe a per-entry bind bit in a namespace/capability list that is checked specifically on a capability-list write operation to permit or deny propagation into another list.

Further, the reference (WO2023067295A1) describes sealed capabilities used as callback tokens and enforced restrictions on use based on trust level, disclosing token-like capability handling and controlled use/propagation semantics. Not disclosed: It does not describe a capability-based lambda-calculus function-application instruction that performs an in-domain call and stores a return address into a machine-status register.

Further, the reference (WO2023203307A1) teaches write-side tracking when a capability is written into a memory region (setting a tracking field), aligning with monitoring/controlling capability propagation via stores. Not disclosed: It does not describe a propagation-permit/bind bit that is checked at the time of writing a capability into a distinct capability-list namespace to allow or deny copying into that list.

Further, the reference (WO2018215734A1) discloses capability permission-flag interpretation mechanisms enabling additional permission semantics from existing capability bits, consistent with implementing a dedicated propagation-control permission within capability metadata. Not disclosed: It does not describe a specific “bind/non-propagate” bit enforced on capability-list write operations to prevent propagation between capability lists or namespaces.

Further, the reference (WO2017182770A1) teaches explicit operations that access and modify capability metadata (including setting/clearing metadata identifying a data block as a capability), relevant to write-path validation/marketing of capability-bearing storage. Not disclosed: It does not describe a dual read-gate/write-gate mechanism that validates a capability token’s internal fields on every capability-list read and capability-list write.

Further, the reference (Reference 1) defines permission-checked capability loads/stores (including constraints on storing capabilities) and defines CJALR as a capability jump-and-link primitive that records a return capability, matching dual-path validation and capability-based call/link behavior. Not disclosed: It does not describe a lambda-calculus-named instruction that performs function application while storing the return address specifically in a machine-status register.

In order for your patent application to be approved, your invention must meet three main criteria established by the USPTO. Your invention must be novel, non-obvious, and useful.

Regarding novelty, we believe there is a good chance the patent office will consider your invention novel.

This brings up the second criteria: your invention must be non-obvious. What this means is that your invention must not be easily perceived by a person of expertise in that invention's particular field. For example, one would not be able to patent a machete duct taped to a rifle or a toaster that can toast ten pieces of bread at once because those inventions would be considered obvious. As well, one would not be able to patent a toaster for the bedroom (as opposed to the kitchen), a bandana that gets tied around a dog's neck (as opposed to a human's), nor any existing inventions/technology used in a different location for the same or different purpose. With that being said, we believe there is a good chance the patent office will consider your invention non-obvious. In order to draw this conclusion, we must ask several questions. For example, would the inventors of any of the prior art be able to come up with your invention? How about an expert in the industry who has also seen the prior art? Would a patent officer look at the most relevant pieces of prior art and think your invention is too similar to those? Would a patent officer think that you are using an invention that already exists and applying it in a different scenario or setting?

If the answer to any or all of these questions is, "yes", your invention would likely be considered obvious or not non-obvious, but we said, "maybe". In reality, the answers to these questions are subjective and truly depend on the patent examiner when deciding if your invention is patentable. As mentioned above, using an existing technology and applying it to a different setting, scenario, or situation is considered an obvious interpretation of the prior art. Once again, the answers to these questions are subjective and ultimately up to the patent examiner. With that being said, we believe there is a good chance the patent office will consider your invention non-obvious.

The best way to use the information in this search report is to determine if you can work around the prior art. What else can you add to the invention to make it more different from the prior art? The concept is to add features and uniqueness to your invention so that it is as different as possible from the prior art. Any changes you can make to differentiate from the prior art will greatly increase the chances of eventual patent success and increase the value of your patent application.

Once you have concluded that you have maximized the uniqueness of your invention in light of the prior art we have mentioned above, let your invention consultant know you are ready to proceed forward with a patent application.

If you decide to move forward with a provisional patent application, we strongly recommend you select our Professional level package which includes professional writing and drawings of your patent application details. Our engineers will be able to fully elaborate on the unique aspects of your invention in light of the prior art. The key is to fully detail all the unique aspects. Our engineers can ensure the unique features are highlighted and detailed to the fullest extent to maximize the value of your provisional patent application.

It is important to note that in light of recent changes in patent law, method patent applications have an added risk of rejection if considered to be an unpatentable method. For your invention, if the examiner believes your invention is focused on financial transactions, human behavior, or using a computer for generic computer functions, the examiner can reject the method patent application. However, because your invention is not focused on purely human behavior and involves a computer aspect, your invention may have a good chance of being considered a patentable method, although the risk of rejection nonetheless exists.

Contact your consultant to discuss the search results in more detail or to proceed with the patent process.

RECOMMENDED STEPS FOR UTILITY PATENT

If you decide to file a utility patent application,

Decide your path

PROVISIONAL UTILITY PATENT APPLICATION

12 months temporary. Option to convert to a non-provisional later, within 12 months.

Provisional Patent Application Basic Package

Provisional Patent Application Essentials Package:

+ Professional Technical Writing **OR**
+ Professional Patent Drawings

Provisional Patent Application Professional Package:

+ Professional Technical Writing **AND**
+ Professional Patent Drawings

Within 12 months

Non-Provisional Patent Application

All parts required by the Patent Office,
including technical writing, patent
drawings, patent claims.

NON-PROVISIONAL UTILITY PATENT APPLICATION

20 years of patent rights, if approved

Non-Provisional Patent Application

All parts required by the Patent Office,
including technical writing, patent
drawings, patent claims.

Recommended
for you

Recommended
for you

RECOMMENDATION NOTES

Your invention as currently disclosed has an overall likely good chance of patent success. Whether or not you decide to add new features to your invention, it is extremely beneficial to have our engineers write technical descriptions which detail how all the features work and how they are different from the prior art. While you did provide a brief explanation of your invention, it is our recommendation that you add several more drawings, descriptions, and details to your application. In particular, you should provide as much information as possible when describing each feature and system component that your invention comprises. These drawings and specifications are very important and can be professionally done by our engineers. They are important as they help show and define the components of your invention, as well as how they work together in a clear manner. It is important to highlight every aspect of how your invention works. We want to explain exactly what makes your invention both novel and non-obvious in respect to the prior art. This is done in our Professional Package provisional patent application service. Let your invention consultant know you want to proceed with either the provisional or non-provisional patent application.

PROVISIONAL OR NON-PROVISIONAL? HOW TO DECIDE

Decide your path

PROVISIONAL UTILITY PATENT APPLICATION

12 months temporary. Option to convert to a non-provisional later, within 12 months.

COMMON SITUATIONS:

- 1) You are not ready to financially commit to the higher cost associated with a non-provisional and you want 12 months to make the decision or to gather funds.
- 2) There is a high chance you will make major changes to your invention within the next 12 months.

NON-PROVISIONAL UTILITY PATENT APPLICATION

20 years of patent rights, if approved

COMMON SITUATIONS:

- 1) You are ready to financially commit to the higher cost associated with a non-provisional patent application.
- 2) Your invention is generally finalized and there is a low chance of major changes.

Use the provisional patent application if you need 12 months of time to decide if you will commit to a non-provisional patent application. You can use the 12 months to raise funds, test out the market, get feedback, improve upon the invention, or any other way to *gain confidence* before committing the additional costs to a non-provisional patent application.

However, if you don't need 12 months of time to make the decision to file a non-provisional, you can skip the provisional application and go straight for the non-provisional. Generally, you go directly to the non-provisional application if you meet two conditions. First, you are financially ready and willing to commit the larger investment, as the non-provisional application is always more expensive than the provisional. Second, you do not anticipate making changes to your invention. Once a patent application is filed, it cannot be changed. If you make changes to your invention, you would have to file a brand new patent application. Most inventors want to make sure that when they do invest in filing a non-provisional patent application, the invention is generally finalized. If you make major changes to the invention after filing a non-provisional, you would have to file a second non-provisional patent application to include those new changes.

RECOMMENDED STEPS FOR DESIGN PATENT

Remember, a design patent application works in parallel with a utility patent application. Design patents offer additional protection for the exterior shape of your invention only. It does not protect functionality.

Your invention is considered unsuitable for a design patent.